Anshuman Rupesh Agrawal

AI/ML and software developer · deep learning for medical imaging, .NET and MERN, hardware prototyping

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Summary

Third-year B.E. Information Technology student at SSGMCE Shegaon and an AI/ML Developer Intern at One Smarter, Inc. I work across deep learning for medical imaging, ASP.NET Core and MERN applications, and hardware prototyping, and co-authored an IEEE-format paper on multiclass skin lesion classification with EfficientNetB3, focal loss and Grad-CAM. Comfortable across the whole lifecycle, and as interested in what a model gets wrong as in what it gets right.

Experience

Internships and roles

AI/ML Developer Intern

Aug 2026 – Present

One Smarter, Inc. · Ohio, remote

- Healthcare data automation: transforming EDI transaction data into structured formats and generating reports without manual handling.
- First system I have written where handling sensitive data set the constraints rather than the features. Code is proprietary and stays private.

Chairperson, Robotics and Automation Society

Jul 2026 – Present

SSGMCE Shegaon

- On the organising team for ICICGR 2026, the IEEE International Conference on Innovations in Communication, Geoscience and Robotics, coordinating technical sessions and hybrid presentations.

AI/ML Intern

Jun – Jul 2026

IEEE EMBS Pune Chapter

- Model training, system architecture and application development for DermAI Monitor, under the mentorship of Dr. Ankita Tidake.
- Co-authored the accompanying IEEE-format paper covering methodology, results and documented limitations.

MERN Stack Development Intern

Aug – Nov 2025

Webstack Academy, Bangalore

- Built Homely Hub end to end across an SDLC-structured cycle, from foundational HTML, CSS, JavaScript and React through to a deployed full-stack application.

Skills

What I can be handed on day one

Machine learning	TensorFlow, Keras, scikit-learn, transfer learning, focal loss, Grad-CAM, model evaluation, computer vision
Generative AI	OpenAI API, prompt engineering
Languages	Python, C#, JavaScript, Java, C, SQL
Web and software	ASP.NET Core MVC, Entity Framework Core, React, Node.js, Express, Flask, REST APIs
Data	MongoDB, MySQL, SQLite, pandas, NumPy, Matplotlib, Chart.js
Tools	Git, GitHub, Linux, AWS fundamentals, Streamlit, Visual Studio, Jupyter, Google Colab, Fusion 360

Selected projects

Newest first

DermAI Monitor

2026

Deep learning system for skin lesion classification and monitoring

- Seven-class dermoscopic classifier on an EfficientNetB3 backbone, trained on a merged, deduplicated HAM10000 and ISIC 2019 set. Two-phase transfer learning, focal loss, class weighting and targeted oversampling cut a 58× class imbalance to roughly 28×.
- Grad-CAM explains what drove each prediction and a classical ABCD dermoscopic score cross-checks it independently. Shipped as a Streamlit application with SQLite-backed longitudinal patient history and automated PDF reporting via ReportLab.
- At 0.5426 macro F1 this is a triage aid, not a diagnostic tool. Documented as an open problem in the paper rather than hidden.

68.97% test accuracy | **0.9143** macro ROC-AUC | **15,615** images | **7** classes

TensorFlow · EfficientNetB3 · Grad-CAM · Streamlit · SQLite

Expense Budget Management

2026

ASP.NET Core MVC on .NET 10 · One Smarter, Inc.

- Income, expense and per-category monthly budget tracking, built over four weekly iterations. Three layers held strictly apart: controllers never build queries, services hold every rule, EF Core sits underneath.
- Every query filters on the user id from the auth cookie rather than form input, so a changed id in a URL returns not-found. Money is decimal(18,2), never float — SQLite has no decimal type, so services total the period's amounts in C# rather than rounding in SQL.

41/41 smoke tests pass | **~2,800** lines of C#

C# · ASP.NET Core MVC · EF Core · SQLite · Chart.js

Homely Hub

2025

Full-stack property rental platform · Webstack Academy

- Serves both tenants and owners: JWT authentication, listings with filtered search across price, location and amenities, an admin dashboard and image uploads. Frontend on Vercel, API on Render, data in MongoDB Atlas.

React · Node.js · Express · MongoDB · JWT

Flashy Learn

2025

AI flashcard generator · Innovo Hackathon, SSGMCE

- Flask application generating study flashcards from a topic prompt through the OpenAI API, for students without access to prepared material. Decks persist so a learner can come back to them. Built with a team of four under hackathon deadline.

Python · Flask · OpenAI API · SQLite

Education

B.E. Information Technology

Sep 2024 – Jul 2028

Shri Sant Gajanan Maharaj College of Engineering, Shegaon

CGPA 8.57 across the first three semesters. HSC and SSC at School of Scholars, 2024 and 2022.

Certificates and training

CS50P: Introduction to Programming with Python
Harvard University · nine problem sets, final project · 2026

AWS Academy Graduate, Cloud Foundations
Amazon Web Services · 20 hours, Credly verified · 2026

Red Hat Linux Fundamentals, RH104
Red Hat · system navigation and command line · 2026

AI and Machine Learning industrial training
iBase Electrosoft with SSGMCE · 10 days · 2026

Generative AI workshop
NxtWave · first end-to-end AI model · 2025

Digital Fablab completion
SSGMCE · all eight modules, later a mentor there · 2025

Hardware, and away from the screen

Three builds from the IEEE SSGMCE Prototype to Product series: an HC-SR04 ultrasonic monitor on an ESP8266 serving its own page of live readings, built at IIT Bombay; a Raspberry Pi infrared presence detector; and ESP32 interfacing at Pursuit 2026. All eight SSGMCE Digital Fablab modules completed, mentoring there since January 2026. State-level softball; district level in cricket and chess.